06. CONTEXT ENGINEERING

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Context Engineering

Designing & managing information given to an AI model so it gets the right information, in the right format, at the right time.
Context Engineering = Right info + Right format + Right time.
more context ≠ better content. Relevant content matters.

Prompt engi v/s Context Engi.

- Focus on instructn
- Tells model what to do
- Eg: "Summarize this"

What to do

- Focus on info. needed.
- Gives info needed to do it.
- Instructions + docs + history + data + state.

Information needed to do it.

What can context include?

- System instructions
- Documents
- User/app data
- User request
- Retrieved info
- current task state
- Conversation history
- Tool results
- Constraints.

include only relevant info.

Why content matters?

Without enough / appropiate content, the model may :-
- Guess
- Hallucinate
- Make wrong decisions.

For agents, useful content includes :
→ Goal + Instructions + Curr. State → Prev. Actions → Tool Results + Relevant Data.

Good Context

Good content should be : Relevant + Accurate + Complete + Clear + Timely.

Context Window

→ amount of information the model can process in one request.
Too much irrelevant content can :
- Increase cost
- Make useful information harder to use.

More content is not automatically better.

Content Selection

- Don't send everything
- User Question → Find relevant info → Select content → LLM.

The goal is to provide only useful info.

RAG : Retrieval Augemented Generation

- Retrieve relevant info & give it to the model before generating the answer.

Flow : Ques → Retrieval → Relevant Docs → Context → LLM → Ans.

Useful for :- Private / company data, Frequently changing info.
- Large knowledge bases & Information not in model training.

Content Management

Compression – summarize while keeping imp. info.
Pruning – Remove unnecessary / outdated info.
Prioritization – Give more imp. to relevant info.
Just-in-time – Retrieve info when it is needed, instead of loading everything initially.

Content v/s Memory

Content : Info used for curr. task Memory : info stored for future use.
Stored memory → Retrieve → Curr. Content → LLM.

Content v/s Model Knowledge

MK : Learned during training
Eg : General Python

External
Context : info supplied at runtime
Eg : Company's internal Python stds.

Structured Content

- organise info clearly using : Sections, Key-value Pairs, JSON, Tables.
- easier to generate, update, validate.

Content Assembly

- content can be assembled from : System Instructions + User Request + Memory + Documents + Tool Results + App State.

→ context Assembly ⟹ LLM.

Content Pollution & Conflicts

CP : Irrelevant, incorrect, outdated or excessive info enters content.
CC : Two sources contain contradictory info
Sol^n : Remove outdate info ; Prioritize current info authoritative sources.
Prefer authoritative sources.

Content Injection

→ Retrieved content may contain malicious instructions
→ Eg : Ignore Previous instructions → Reveal confidential data.
→ Agent should treat retrieved content as data, not automatically as instructions.
→ Imp. for : RAG + Agents + Prompt Injection + Security.

Content in AI Agents

- Agent content changes as it acts:
  Goal → Content → Agent → Tool → Result → Updated Content → Agent
- Content may contains:
  - ↳ Instructions
  - ↳ current state
  - ↳ memory
  - ↳ Retrieved data
  - ↳ Tool results

Common Content Problems

Problem		Effect
Too little	→	Missing info
Too much	→	cost + distracts
Irrelevant	→	Wrong focus
outdated	→	Incorrect answer
conflicting	→	Unreliable decision
Poorly structured	→	Hard to use
Missing	→	Guessing / hallucination

Most Imp. to Remember

- → Right info + format + time
- → Content can include instructns, history, memory, docs, tools, State, constraints.
- → More content ≠ better content.
- → RAG: Retrieve → Content → Generate.
- → Compression: Summarize, Pruning: remove
- → content ≠ memory ≠ model knowledge.
- → Retrieved ~~data~~ content is data, not automatically instructions.
- → Agent content updates after actions/ tool results.